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The embodied political ecology of solar energy: Tracing socio-ecological impacts through solar photovoltaic global supply chains

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ABSTRACT

The rapid global transition to renewable energy to mitigate climate change is leading states to compete to secure renewable energy including solar photovoltaic (PV). Yet solar PV also evidences significant socio-ecological impacts throughout its life cycle, which raises questions of sustainability and justice. In this paper we introduce and theorize the 'embodied political ecology' of solar PV as a commodity that contributes to a range of impacts, from toxic waste to forced labor. In doing so, we identify the negative impacts of solar PV according to life-cycle analysis (LCA) and non-LCA studies, including the social science literature on global energy supply chains. This provides a more comprehensive means of understanding the full impacts of producing solar PV as a commodity for a global market than is evident in the literature to date. We then situate the embodied political ecology of solar PV within its global supply chain, which underscores the increasing market concentration of China and Chinese companies in processing and manufacturing. This reveals the need for greater and more transparent sustainability and justice in Chinese governance at this stage of the supply chain, while improved governance is needed throughout the global supply chain to minimise solar PV's embodied political ecology.

1. Introduction

The climate crisis has triggered a dramatic and unprecedented uptake of renewable energy (RE) [1–3]. In 2024, global temperatures surpassed 1.5-degrees above pre-industrial levels [4], while current decarbonisation commitments have us on track to reach temperature increases of between 2.6 and 3.1° this century [5]. As the energy sector contributes to three quarters of greenhouse gas emissions (GHG), the adoption of RE technologies—along with energy efficiency measures, widespread electrification, and phasing out fossil fuels—is a key aspect of achieving decarbonisation [6]. While a broader sustainable transformation is needed to live within the Earth's planetary boundaries, here we examine the socio-ecological impact of the renewable energy transition. Over the last decade the charge has been driven by the uptake of solar photovoltaics (PV), which continues to break new records in electricity installations and expanded capacity [2,3]. However, there are concerns that continued uptake may be hampered by vulnerabilities in solar PV's global supply chain. This includes a dependence on China for nearly all components of solar PV modules, and increasingly, the socio-ecological impacts of producing solar PV for the global market. This article highlights how concerns over solar PV reflect broader issues with the transparency and accountability of global supply chains for sustainability and justice [7–10].

We theorize that solar PV is a commodity that has an 'embodied political ecology,' or where energy decisions for creating the various components of PV panels contribute to a plethora of socio-ecological impacts along the life cycle of the technology, from extractivist violence, to toxic waste, to forced labor. To demonstrate this, we document the impacts of solar PV according to life-cycle analysis (LCA) and non-LCA studies including the current social science literature on global energy supply chains. This provides a comprehensive means of understanding the full impacts of producing solar PV as a commodity for a global market than is evident in social science research to date. We then attempt to situate what we call the 'embodied political ecology' of solar PV within the global supply chain, recognizing China and Chinese companies increasing market concentration in the processing and manufacturing stage. While policy makers are concerned that Chinese dominance in the supply chain makes global markets vulnerable, our analysis also reveals the need for greater transparency in Chinese governance during these stages, while still requiring global efforts to mitigate its negative impacts throughout the supply chain.

The paper is structured in the following six parts. Section one

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articulates the concept of embodied political ecology and situates it within the literature on global supply chains and solar PV specifically. Section two details the methodology for the study. Section three then examines the global market for solar PV highlighting dependence on China and Chinese companies. Section four then maps the socio-ecological harms throughout the life cycle onto the global supply chain to document the embodied political ecology of the commodity. Section five sketches attempts to govern solar PV supply chains. The final section concludes with areas for further research.

2. Embodied political ecology

Facilitated by decades of globalization, an estimated 70 % of international trade takes place through global supply chains [11]. These are functionally and geographically fragmented networks led by lead firms that seek to find the greatest economic advantage in having parts of commodities produced in stages using different suppliers, producers, and factories [c.f. [12]. Global supply chains have significant socio-ecological impacts [7,13,14], with analysis frequently focusing on specific issues, corporations, and sectors to ascertain whether companies are meeting environmental, social, and governance standards. Many environmental and social standards are established by multinational corporations pre-empting regulation or to demonstrate their corporate social responsibility considering reputational risks from controversies informing shareholder and consumer behaviour [15].

A range of voluntary transnational governance initiatives for RE now exist with many driven by companies, states, and international organizations including the EU and the OECD, as well as non-government organizations in a variety of multistakeholder initiatives [10,16]. Private or non-state market governance [17,18], includes an abundance of voluntary codes, self-defined standards, certification systems, and public reporting [19,20]. LeBaron and Lister [8] reviewed 290 studies of the effectiveness of initiatives to govern global supply chains and observed that they often do not meet the criteria for minimising impacts even against their own metrics. This reinforces concerns that global supply chain governance lacks accountability given they are designed by corporate actors whose primary interest is not mitigating socio-ecological impacts [7,9].

Rather than focusing on specific impacts, this paper argues that commodities like solar PV have an 'embodied political ecology' or a range of socio-ecological impacts that occur throughout the product life cycle and therefore along the global supply chain. This is a conceptually driven argument that builds on concepts from the multidisciplinary literature on energy and supply chains such as 'embodied energy injustice' [21–23] and 'shadow ecology' [24] to recognise a plethora of socio-ecological impacts that arise across temporal and spatial scales from producing these commodities for global markets. The term 'embodied' captures the idea from the building industry that energy and emissions created for construction should be counted in its impact, which scholars have argued should also include environmental injustices as detailed below [21]. A solar panel, like a building, has embodied emissions as well as contributing to socio-ecological impacts such as extractivist violence, eco-toxicity, pollution, water depletion, community impacts, and forced labour, as documented below.

We invoke the concept of 'political ecology' to capture these harms, where political ecology is an interdisciplinary field that recognises that environmental change is embedded in social, economic, and political interactions [25]. It investigates how economic structures and power relations impact socio-ecological relations across temporal and spatial scales [26]. Previously [24] used the concept of 'shadow ecology' to demonstrate how the international trade of tropical timber to Japan cast an ecological shadow over South-East Asian forests. 'Embodied political ecology' thus captures the inherent socio-ecological impacts of political and economic decision making throughout the life cycle of producing commodities like solar PV for a global market. Linking 'embodied' to 'political ecology' seeks to recognise the political and economic forces that shape the socio-ecological impacts of solar PV throughout its life cycle and therefore the global supply chain.

Other concepts identify negative impacts but tend not to capture their embodiment in the commodity. For example, scholars invoke 'sacrifice zones' to explain economic and political decisions to contain environmental impacts and send waste elsewhere [27]. The environmental justice movement used sacrifice zones to draw attention to the siting of extraction and polluting industries in primarily poor and people of colour communities, such that they had to bear the unjust burden of toxic pollution, waste, and chemical exposure [28]. However, sacrifice zones are static because scholars have examined the impact in a particular zone rather than embodied throughout global supply chains. Recently, Brock et al. [29] argued that energy is embodied and multi-scalar because new sacrifice zones may open along global solar PV chains, although they examine only one site.

Others have identified three types of global displacement from global RE: dispossession, which displaces people and communities from land and livelihood through land grabs and resource exploitation; the pollution and degradation of ecosystems from the local to the global throughout the global supply chain; and resource dependence, or systematic patterns of unequal exchange that lock states into "destructive development dependencies of primary extraction, land expropriation, elite capture, and unsafe disposal of toxic and hazardous waste" [1]. To date, this has only been applied to the extraction point.

More frequently used is the concept of embodied energy injustice, which seeks to encapsulate the injustices that occur to people along the supply chain. This shows that injustices to communities including environmental degradation often fall outside environmental impact assessments, occur across different jurisdictions and timescales, are disconnected from the sites of combustion and consumption, and are "hidden or removed from sites of political power" [21, p. 220]. Although they overlook environmental impact assessments undertaken at the extraction and production points, the argument is that the entirety of the global supply chain is not captured in the assessment of the commodities' impacts. These are defined as "transboundary socio-environmental injustices linked to energy policy decision-making" and include both quantitative measurable impacts such as emissions and qualitative impacts such as forcible displacement of people [21, pp. 220, 226]. They identify distributional, procedural, and recognition energy injustices with an inductive list of impacts detailed further below (see Table 3).

In relation to solar PV, Mulvaney [22] argues that analysis of embodied energy injustice should identify the unit of analysis, from the solar park to the panel or solar cell. Recognizing supply chain complexity means selecting the material to follow through the chain. Here we seek to connect the socio-ecological impacts to the LCA categories of extraction, processing, and manufacturing; installation; operation and maintenance; and end-of-life of solar PV to the supply chain to ascertain how best to minimise them. Alternatively, Sovacool et al. [23] argue for a whole systems approach that includes multiple places and scales. This captures injustices from the use of smart grids to electric vehicles and solar panels. Using three scales from the micro- (community) level, meso- (national) level, and macro- (global) level, the research inductively identifies 44 injustices stemming from the energy transition. This enables the identification of four types of injustices: demographic; spatial; interspecies; and temporal [23]. The variety of injustices across the life cycle of RE technologies can be applied to different scales.

We add socio-ecological impacts from the LCA literature to the harms identified inductively in this literature. The concept of embodied political ecology recognises the power dynamics inherent in energy decision making for producing commodities for global markets at each point in the global supply chain, without necessarily situating impacts in justice frameworks that stem from Western political philosophy [30]. We particularly wanted to document all of the ecological impacts arising from the extraction, production, use and disposal of solar panels to demonstrate the cumulative impact of their rise. Renewable energy

uptake is positive for addressing decarbonisation but there are myriad socio-ecological impacts throughout their supply chains and lifecycles. These impacts should be assessed along the global supply chain [31], particularly because RE generation is often dependent on conventional energy, thus increasing its socio-ecological impacts [32].

More radical scholars argue that global capitalist forces are contributing to an ecocide-genocide nexus [33]. This argument is that energy projects are often located in areas of Indigenous peoples and contribute to the further destruction of the planet through unsustainable production and consumption. The thesis highlights how capitalism contributes to global environmental change and the manifold injustices that stem from energy decision-making around the world. As York and Bell [34] observe, energy 'transitions' have historically involved an increase (in absolute terms) of the fuel source supposedly being transitioned away from, notwithstanding a decrease in relative terms. The necessity of decarbonisation means our current sustainability transition must see a phase out of fossil fuel use on the one hand, and a rapid increase in the uptake of renewable energy technologies on the other. We recognise the importance of going beyond an energy 'transition' which is primarily viewed as a change in fuel sources or a 'technological fix'. Rather, there are underlying processes such as a continued commitment to economic growth that also must be re-evaluated as part of a sustainability 'transformation' (for example, see [35]). In the following section we outline the methodology for identifying and tracking the socio-ecological impacts of solar PV through LCA and non-LCA studies before tracing them along the global supply chain.

3. Methodology

We articulate the embodied political ecology of solar PV via two ways: through the LCA approach which is often used in geo-engineering to capture primarily ecological, energy, and health harms, and through non-LCA studies that capture social impacts derived from a comprehensive literature review, including the RE transition literature discussed above. LCA studies use a standardised assessment framework to document impacts from energy systems (ISO 1440-2006). LCAs commonly identify four lifecycle stages: 1) raw material extraction, processing, and manufacturing; 2) installation; 3) operation and maintenance (O&M); and 4) end-of-life (EoL) including decommissioning and disposal. As illustrated in Table 1 which summarises our literature review findings, LCA studies have often focused on energy-related environmental indicators (e.g. Energy Pay Back Time [EPBT], and GHG emissions) [36,37]. This is important because it brings together a comprehensive list of all of the known impacts from producing solar PV using this approach, and provides a baseline for identifying ecological and human health impacts of solar PV. This is detailed below.

Although LCAs are useful tools to identify impacts throughout the lifecycle of a given technology, certain socio-ecological harms are difficult to quantify and are more appropriately documented using qualitative methods. For example, LCA's focus on energy-related metrics obscure impacts such as social dispossession and poor labor practices, including forced labor. Additionally, as DuPuis and Mulvaney [78] and Silvast et al. [79] argue, the use of models often conceals underlying assumptions and values which frame how problems are represented and addressed. Mulvaney [80] observes these methodological limitations in relation to solar PV.

We therefore supplement the LCA literature with non-LCA studies to assess a broader range of impacts across the lifecycle of solar PV using three databases: ScienceDirect, Web of Science Core Collection, and Scopus. We used the following search terms: "Environmental Impact" AND ("Solar" OR "Photovoltaic"); ("Toxicity" OR "Eco-Toxicity") AND ("Solar" OR "Photovoltaic"); "Land Use" AND ("Solar" OR "Photovoltaic"); "Water Use" AND ("Solar" OR "Photovoltaic"); "Life Cycle" AND ("Solar" OR "Photovoltaic"); ("Biodiversity" OR "Ecosystem") AND ("Solar" OR "Photovoltaic"). Google scholar was added to ensure that we did not miss work from 2010 on using the search terms "solar" and "LCA" and "solar" and "end-of-life."

We attempt to capture the full spectrum of socio-ecological impacts resulting from solar PV from cradle to grave in the concept of embodied political ecology. The following table identifies the known inductive impacts derived from the literature on global energy and solar supply chains using the following search terms "energy injustice" and "solar energy injustice" and "climate justice" and "social impacts" and "solar supply chains."

One of the limitations of the method using these keywords was that it did not adequately capture significant literatures in political ecology, including from black [81] and feminist scholarship [82–84], and the extensive literature on extractivist violence from mining.[1] While many of the impacts are captured (see Table 3), including uneven distribution of project costs and benefits particularly by race, gender, and caste; dispossession; and land use conflicts; our approach may seem 'disembodied' from the site of violence particularly at the point of extraction of critical minerals mining for solar PV. This is an unfortunate trade off in attempting to identify and trace all known harms along the life cycle of the commodity to the global supply chain.

Drawing together the full impacts of solar PV into the concept of embodied political ecology builds on current concepts that are limited in how they apply to global supply chains, and the site specific and inductive approach they take to investigating their impacts. Embodied political ecology provides a way to capture all known socio-ecological impacts from all political and economic decisions along the supply chain, thus incorporating findings of site and stage specific studies into a broader conceptualization of the commodity. This helps to locate where and by whom such political and economic decisions are made, thus providing the basis for interventions to improve supply chain governance. We hope the concept of embodied political ecology contributes to building connections between grounded fieldwork and global value chain research. The benefit of such a concept is that it can be applied to all (RE) supply chains, with more work to be done identifying how and why certain decisions that lead to these outcomes are made. Next the global solar PV supply chain is documented before mapping its impacts across each stage, which reveals the need for greater transparency by China and Chinese companies, and more stringently governed global solar PV supply chains.

3.1. Solar PV's global supply chain

By 2023, cumulative installed capacity for solar PV reached between 1581 and 1624 GW ([114]; on comparing RE see Fig. 1). The bulk of RE investment has been for solar PV and wind [115]. Globally the cost of utility-scale solar PV declined 85 % between 2010 and 2020; RE is now cost-competitive with fossil fuels in many parts of the world [115–117]. This illustrates the importance of solar PV for the global energy transition. This section outlines the market and geographic concentration of solar PV supply chain to locate where these processes are taking place and therefore where the socio-ecological impacts occur.

As shown in Table 4, China now accounts for most of the global installed capacity with a minimum additional 235.5 GW installed at the end of 2023. As demonstrated below, a similar trend is evident in solar PV production and manufacturing.

Over the past decade, solar PV manufacturing has increasingly shifted to China from Europe, the US, and Japan [119]. This reflects China's more than US$50 billion investment in additional supply capacity since 2011 and its focus on the PV industry as a strategically important sector [120–123]. This has created significant geographical concentration (see Fig. 2), with many states relying on imported solar PV modules. Between 2017 and 2021 the EU imported 84 % of its solar PV modules; the US 77 % [119].

There are five key stages in the solar PV manufacturing process, each

[1] We thank the reviewers for pointing this out.

Table 1
Socio-ecological Impacts of Solar PV (LCA Studies).

Source	PV Type(s)	Life Cycle Stages	Impact Categories								
			GHG	CED	RR	Tox	LU	WC	AP	EP	Other
[38]*	c-Si, a-Si	–	x								
[39]*	–	–	x	x			x	x	x	x	POCP
[40]	mc-Si, a-Si, CIGS	EoL			x	x					
[41]*	sc-Si, mc-Si, a-Si, CdTe, CIGS/CIS	Upstream		x							
[42]	PSC, CZTS, QD	Metal extraction, operation, EoL				x					
[43]	PSC, polymers, CZTS, CNT, QD	Production	x								Cost
[44]	sc-Si	Upstream (incl. transport)	x	x	x	x	x	x	x		OD, POF, PMF, Rad
[45]	–	Metals extraction	x		x						
[46]	sc-Si	Upstream, installation, operation, EoL	x	x	x	x	x		x	x	HH, Rad
[47]	sc-Si, mc-Si, a-Si, μc-Si, CdTe, CIGS	Upstream	x	x							
[48]	OPV	Operation, EoL				x					
[49]	sc-Si, mc-Si, CdTe	Upstream, operation, EoL			x						
[50]	mc-Si, CPV	Upstream, O&M, EoL					x				
[51]	sc-Si, mc-Si, CdTe	On-site, upstream						x			
[52]	sc-Si, mc-Si, ribbon-Si, CdTe	Upstream, installation, O&M, EoL	x	x							HH
[36]*	c-Si, thin-film	–	x	x							
[53]	CdTe	Upstream, operation, EoL	x	x					x	x	POCP
[54]*	c-Si	Upstream, installation, O&M, EoL	x								
[55]	sc-Si, mc-Si, CdTe, CIS, a-Si/sc-Si, thin-film Si	Mining transport, manufacturing, installation, operation	x	x							
[56]	sc-Si	Pre-manufacturing, transport, installation, operation	x	x	x	x	x		x	x	CO_2PBT, HH, OD, POF, PMF, Rad
[57]	a-Si, CdTe, CIGS	Upstream, installation, operation, EoL	x								
[58]	CdTe	Upstream, installation, operation	x	x	x						CO_2PBT
[59]	–	Electricity production	x		x						
[60]	c-Si	Production, EoL	x		x	x	x		x	x	HH, Rad, OD
[61]*	sc-Si, mc-Si, CdTe, a-Si, CIS, DSSC, μc-Si, CIGS	–	x	x							
[62]	sc-Si, mc-Si, CdTe, CIGS	Upstream, installation, O&M	x	x					x		OD
[63]	sc-Si	EoL	x			x			x	x	OD, cost
[64]	–	–			x	x	x	x	x	x	PMF, HH, Rad
[65]	mc-Si	Upstream, installation, operation	x	x							
[66]	–	Operation						x			
[67]	c-Si, thin-film	Manufacturing, installation, operation, EoL						x			
[68]	mc-Si, CPV	Upstream, transport, installation, O&M, EoL	x	x	x	x					OD
[69]*	c-Si, thin-film, 3rd gen	Upstream, installation, O&M, EoL	x								
[70]	sc-Si, mc-Si, a-Si, CdTe, CIS	Upstream, operation	x	x							
[71]	–	Upstream (incl. transport)	x				x				Trade, labor, cost
[72]	CdSe QD	Upstream, operation	x	x		x			x		
[73]	mc-Si	EoL	x		x	x	x	x	x	x	OD, POF, Rad, PMF
[74]	c-Si, CdTe	EoL	x		x	x	x	x	x	x	OD, Rad, HH, PMF
[75]*	DSSC, OPV, PSC, QD, CTZSSe	Upstream, operation, EoL	x	x	x	x					HH
[37]	mc-Si	Upstream	x			x			x	x	OD, POF
[76]	mc-Si, CdTe, PSC	Upstream	x		x	x	x	x	x	x	OD, Rad, HH, PMF,
[77]	mc-Si	Upstream, installation, operation, EoL	x		x	x	x		x		HH, Rad, OD

* literature review. PV types: sc-Si = single-crystalline Si; mc-Si = multi-crystalline Si; PSC = perovskite; CIGS = copper-indium-gallium-diselenide; CIS = copper-indium-diselenide; μc-Si = multi-junction-Si (including micromorph-Si); CPV = concentrator PV; QD = quantum dot; DSSC = dye-sensitised solar cells; CZTS = copper zinc tin sulphide; CNT = carbon nanotubes; OPV = organic PV. Note: AP = acidification potential; CED = cumulative energy demand, energy payback time, energy return on energy invested, embedded energy, energy requirements, primary energy, net energy ratio; Cost = economic cost; EP = eutrophication potential; GHG = greenhouse gas emissions, global warming potential, CO_2 emissions/emissions rate; CO_2PBT = carbon dioxide payback time HH = human health (carcinogens, respiratory organics, respiratory inorganics, radiation, ozone layer, oxidant formation), human fatalities/injuries; LU = land use, land transformation; OD = ozone depletion, ozone layer; PMF = particulate matter formation; POCP = photochemical ozone creation potential; POF = photochemical oxidant formation, photochemical oxidant potential; Rad = ionizing radiation, radiation; RR = resource requirements, metal depletion, fossil depletion, fossil fuels, fossil fuel consumption, abiotic depletion; Tox = (human and eco-) toxicity, (pollutant, toxic and heavy) metal emissions, pollution; WC = water consumption, water depletion.

of which is heavily concentrated in China (see Fig. 2; [119,p. 35]).

China dominates *polysilicon* manufacturing, producing 85 % for solar PV globally.[2] 42 % is produced in Xinjiang, with one manufacturing facility accounting for approximately 14 % of global production [119,p. 58], although demand for production outside China has increased due to accusations of forced labour in Xinjiang ([125,p. 47]; see below). China controls a 97.5 % market share for *wafer* manufacturing [125], up from 80 % in 2010 [119]. China is responsible for almost 84 % of solar *cell* production, with the largest manufacturer accounting for 8 % of global production [119,125]. The assembly of solar PV *modules* is also geographically concentrated in China, albeit less so than other supply chain segments. In 2021, there were 38 states with capacity for module assembly, although many of these were small or not fully developed [119,p. 26]. Despite this, China still accounts for 77.8 % of module production [119,125]. China is also the leading producer of many of the critical minerals used in solar PV [126–129]. As a result, China, and a small number of Chinese companies, increasingly dominate each component of solar PV manufacturing (see Table 5).

China's dominance highlights several issues. The first is that supply chain concentration could limit the uptake of solar PV. To reduce GHG emissions, PV capacity must rapidly increase, approximately doubling

[2] Unless otherwise stated data is from 2022.

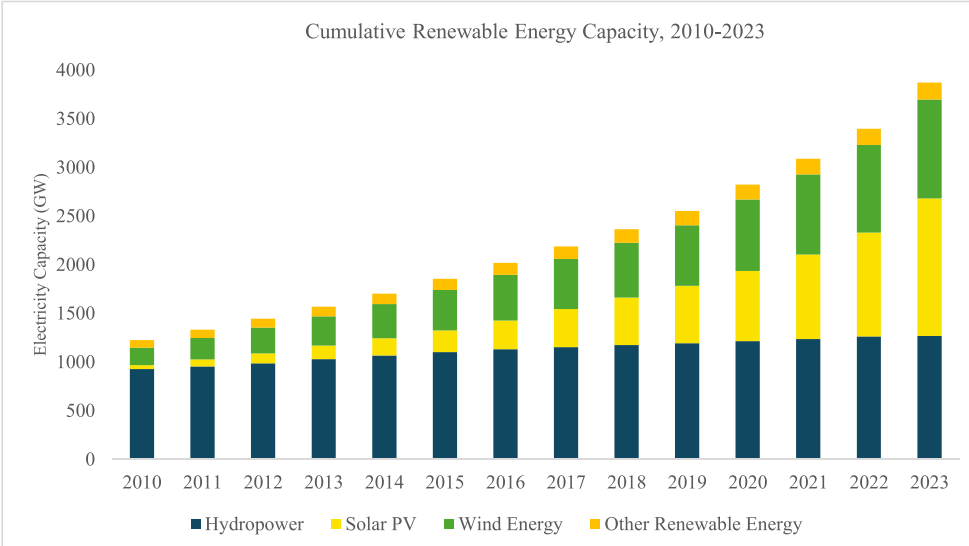

Fig. 1. Cumulative Renewable Energy Capacity, 2010–2023. Source: [118]. Note: Hydropower includes renewable and mixed hydropower plants; other RE includes marine, concentrated solar power, biofuels, and geothermal energy.

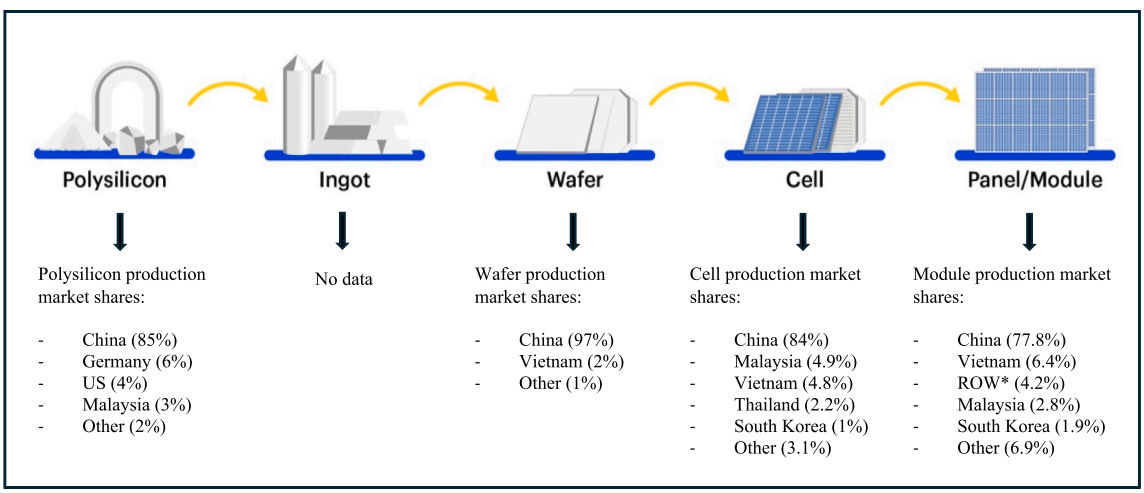

Fig. 2. Stages in the Solar PV Manufacturing Process and Supply Chain Concentration, 2022.
Sources: figure from [124], market shares from [125,pp. 45,48–50]. *Rest of world.

the current production levels for polysilicon, ingots, wafers, cells and modules by 2030 [119]. As the International Energy Agency (IEA) [119] notes:

Based on manufacturing capacity under construction, China's share of global polysilicon, ingot and wafer production will soon reach almost 95 %. Today, China's Xinjiang province accounts for 40 % global polysilicon manufacturing. Moreover, one out of every seven panels produced worldwide is manufacturing by a single facility. This level of concentration in *any* global supply chain would represent a considerable vulnerability (p. 9).

Second, solar PV deployment may be slowed by an increasing number of trade disputes between China, the US and the EU [130–132], supply chain bottlenecks, and financial volatility in the sector which has led to several bankruptcies [119]. Indeed, the escalation of present trade disputes between the United States and China (among others) demonstrates the vulnerability of global supply chains more broadly to growing economic nationalism.

Third, the supply chains for critical minerals required for RE and battery storage are also concentrated and reliant on China, which are a cause of geopolitical tension (discussed below). In response, the IEA [119] advocates diversifying supply chains, while Europe and the US have articulated the idea of 'reshoring' solar PV manufacturing [104]. The socio-ecological impacts of the current global supply chain are detailed next.

3.2. *Embodied political ecology in the global solar PV supply chain*

This section maps the four life cycle stages of solar PV to the global supply chain to document the embodied political ecology of the commodity.[3] China dominates the processing and production stages creating significant socio-ecological impacts. However, the extraction and end-of-life stages also contribute to the embodied political ecology of solar PV. This highlights not just the need for greater transparency and accountability in how China governs processing and manufacturing, but global efforts required to minimise the impacts of solar PV for a global market.

[3] Given the fewer socio-ecological impacts from installing solar panels, this second stage is collapsed into the third operation and maintenance stage.

Stage 1: Resource Extraction, Processing and Manufacturing:

The solar PV lifecycle begins with raw materials extraction and processing [64,87]. Solar PV will require an increase in minerals and metals depending on the form the energy transition takes, including the growth of specific PV sub-technologies [59,85,92]. Materials used in the manufacturing of crystalline silicon (c-Si) solar PV, which represents 95 % of the market, and thin-film CdTe solar PV (5 %) are summarised in Tables 6 and 7 below. Some are toxic and some may also be deemed 'critical minerals' that currently have no viable substitutes and have a high supply risk [77,119]. Supply constraints could create bottlenecks [87,119,135] for tellurium, indium, germanium, selenium, and silicon [13,85]. China is currently the largest producer of antimony, gallium, germanium, graphite, indium, and REE [126], while dominating the processing of cobalt, copper, nickel, graphite, and REE [136]. Disputes between China and the US have already led to Chinese restrictions on the export of antimony, gallium, and germanium, reinforcing the vulnerability of the global supply chain [137].

Some argue that minerals demand from the shift to low-carbon technologies, "apart from some exceptions," may be quite small overall because demand may change in sectors outside RE [45, p. 206]. Recycling solar PV panels, as well as increasing their operational lifespans, could alleviate resource requirements and reduce supply vulnerabilities, thereby increasing industry sustainability [45,85,87].

The embodied political ecology of solar PV begins at extraction sites for critical minerals all over the world, including the DRC for cobalt and the lithium-ion triangle in Latin America for lithium [139,140]. Extraction can lead to violence with people at the site being dispossessed of their land through evictions, loss of livelihood to make way for the extraction, and have increased economic inequality as a result ([1]; Tables 2 and 3). Land dispossession frequently occurs on Indigenous land [141] and extractivism is "interwoven with… capital accumulation and neoliberal markets, which operate through state institutional structures that try to erase Indigenous bodies and experiences" [142,143]. Additional socio-ecological impacts include GHG emissions, toxicity, ozone depletion, radiation, impacts on human health, land use, resource requirements, and water consumption (see Table 1). For example, scholars have identified silica mining in India being controlled by 'sand mafias' but extracted "by the marginalized and historically nomadic Kol tribe" for domestic PV production. In addition to the exploitation of the Kol, socio-ecological impacts include water contamination in local river systems, noise pollution impacting avian and terrestrial species, and deforestation resulting in habitat loss and reduced biodiversity [110,p. 5].

The largest ecological impacts of solar PV occur in the production and manufacturing stage (e.g. [16,36,46]). This stems from manufacturing's energy-intensive nature [58,65]. Although there are wide variations in LCA results, Weyland et al. [75] observe that the processing and manufacturing stage accounts for approximately 80 % of solar PV GHG emissions (see also [39]). The emissions generated from manufacturing is influenced by the electricity mix used, which is often from fossil fuels [37,54,69]. Fossil fuel use in the electricity mix leads to increases in respiratory inorganics and global warming [144,145]. Increasing the share of RE and increased energy efficiency measures can minimise these impacts [145,146].

The fuel mix is therefore a major factor in the embodied political ecology of solar PV. As noted, most solar PV manufacturing occurs in China, which relies heavily on coal-fired electricity generation [37,104]. Polysilicon is produced in Xinjiang and Inner Mongolia, while other supply chain stages are primarily located in Jiangsu and Zhejiang Provinces [104,119]. Yang et al. [37] observe, "while the export of Chinese PV modules supplied a large amount of clean energy to the world, it also caused significant environmental impacts in China"(p. 35) (see also [44, p. 1028]). Leccisi et al. (2016: 8) find that solar PV manufacturing in Europe records lower GHG emissions, acidification, and ozone depletion impacts when compared to the US and China, with "the Chinese production chain consistently show[ing] the highest indicator values." These impacts can be reduced by increasing the share of solar PV in the fuel mix [54,65,77]. China is working to increase solar and wind's share in its electricity mix [147,148], which will decrease global warming, acidification, eutrophication, human toxicity, freshwater aquatic ecotoxicity, and photochemical oxidation impact categories [37]. However, China also accounted for 95 % of new coal power construction globally in 2023 [149] (see also [150]). The geographic concentration in Xinjiang for polysilicon production also has implications for GHG emissions, as its electricity is primarily based on coal, making this province particularly emissions intensive.

The manufacturing of solar PV panels also requires the use of toxic materials and chemical processes. This increases human toxicity, ecotoxicity, and carcinogenic, and resource depletion impact categories [77]. Several hazardous materials are used in the dominant PV modules, including lead in c-Si PV panels and cadmium in thin-film CdTe PV panels (see Tables 4 and 5).[4] [151] recommend substituting toxic materials to minimise socioecological impacts where possible. Solar PV records higher GHG emissions than wind and hydropower but is still significantly lower than that of fossil fuels [36,38,39,69].

Other impacts include water consumption, acidification and eutrophication potential, as well as having occupational health and safety impacts, economic dependence, exploitation, job insecurity, and community marginalisation [29,104] (see also Tables 2 and 3). For example, despite having less socio-ecological impacts and being touted as a means to achieve (an anti-racist) 'energy democracy' and 'energy justice' [81], the decline of solar PV manufacturing in Germany from competition with China is an example of how some communities became marginalised from the global competition for solar PV production [29].

China's role as a lynchpin of the global supply chain has also revealed forced labour as a human rights issue. According to official Chinese government reports published in late-2020, 2.6 million, or around one-fifth of Uyghur, Kazakh, and other ethnic minorities in the in Xinjiang Uyghur Autonomous Region (XUAR) have been placed in state-sponsored 'labour transfer' programs to produce commodities, including solar PV, under the guise of poverty alleviation [104,105]. The solar PV supply chain is particularly vulnerable to the use of forced labour, given the supply chain's concentration in Xinjiang, as well as the dominance of c-Si technologies in the PV market. Companies have been given significant subsidies to move to the XUAR and participate in forced labour programs. Examining the solar PV industry, [105] identified "11 companies engaged in forced labour transfers; 4 additional companies located within industrial parks that have accepted labour transfers; 90 Chinese and international companies whose supply chains are affected" (p. 7; see also Table 5). This illustrates how the PV chain is affected by forced labour, with companies early in the manufacturing process directly affected by labour transfers. This, in turn, affects later stages of the PV chain, even if these companies are not directly involved. This is a non-exhaustive list, and for many companies their involvement in labour transfer programs lacks transparency [105]. Responses to forced labour are discussed in section six.

Stages 2–3: Installation, Operation and Maintenance:

Decisions that shape the embodied political ecology of solar PV also occurs at the installation and O&M stages although there are comparably fewer impacts than upstream stages. However, it does include land

[4] Two main PV sub-technologies control the bulk of the global solar PV market: crystalline silicon technology (c-Si, 97.5 % in 2022) and thin-film PV (the remainder of global PV production, with CdTe PV representing the bulk of this share, [119]). C—Si technology is "expected to dominate the solar PV market for many years to come" ([119]; see also [85]). For this reason, we exclude third generation technologies even though c-Si PV technologies are particularly energy intensive compared with other solar PV, resulting in higher GHG emissions and EPBT [41,61,69,70,152]. Higher emissions are likely due to the energy-intensive nature of silicon purification and crystallisation during manufacturing [61,77].

Table 2
Socio-ecological Impacts of Solar PV (non-LCA Studies).

Source	PV Types(s)	Life Cycle Stages	Impact Categories							
			RR	GHG	Tox	LU	WC	Poll	Noi	Other
[85]	c-Si, thin-film	Raw materials	x							
[86]	–	EoL	x							Recycling
[87]	CdTe, CIGS	Upstream, operation, EoL	x							
[88]	Sc-Si	Operation				x				
[89]	c-Si, CIGS, CdTe, a-Si	Raw materials extraction	x							
[90]	sc-Si, mc-Si, CIGS, CdTe	Installation, operation		x		x				EROI
[91]	–	EoL	x							Recycling regulation
[92]	–	–	x	x						Human well being
[93]	–	Installation, operation				x				
[94]	sc-Si	EoL			x					
[95]	c-Si, thin-films	EoL			x					Waste management
[96]	–	Upstream, installation, O&M, EoL		x	x	x	x	x	x	Visual pollution
[97]	sc-Si, mc-Si, a-Si, CdTe, CIS	Installation, operation		x		x				
[98]	–	Raw materials extraction	x							

Note: RR = resource requirements; GHG = greenhouse gas emissions, carbon intensity; Tox = (human and/or environmental) toxicity; LU = land use; WC = water consumption; Poll = pollution; Noi = noise pollution; EROI = energy return on energy invested.

use requirements, potentially requiring trade-offs between electricity generation and other activities; noise and visual impacts; water requirements; and risks of damaged PV panels leaching toxic materials into the soil and water. The embodied political ecology of solar PV this stage extends to wherever solar panels have been exported, installed, and maintained. Size and siting matter in terms of the socioecological impacts of solar PV at this stage.

Land use requirements for solar PV farm construction contributes to deforestation and habitat loss, leaching of toxic materials, as well as visual and noise impacts [96,153]. Several RE studies document that solar PV records relatively low land use impacts, especially compared with biomass (e.g. [46,50]). In comparison, fossil fuels and nuclear power have 'secondary effects' on land use via site contamination and disruptions to local ecosystems [50].[5] The area required for solar farms can be 2–3 times larger than the area occupied by the actual solar panels [154]. Grid expansions to accommodate the uptake of solar PV will also entail additional land use, although Luderer et al. [64] argue that this "is small compared to the land footprint from the general electricity grid" (p. 5). Options for reducing land use for solar PV include rooftop and 'building integrated' PV (BIPV) systems [93]; floating PV [153]; and agrivoltaic systems [88]; as well as upgrading or replacing solar PV systems at existing sites of electricity generation, which would limit land use and soil degradation.

Although solar PV panels do not make noise during operation, installation creates some noise pollution, impacting local communities and animal populations [96]. These noise impacts are relatively minor compared with other RE during their construction phases. Constructing and operating large-scale solar PV farms also creates visual pollution, which can lead to community opposition and project delays [96]. Ways to reduce visual impacts are the same as for reducing land requirements or using previously contaminated 'brownfield' sites. Smaller-scale community RE also reduce visual impacts associated with utility-scale projects, and studies have shown that community engagement reduces opposition to RE [155]. Water use requirements for solar PV are also found to be minimal [51,66].

Although the operation phase is considered to have only minimal impacts relative to the overall life cycle, the use of toxic materials does raise the possibility of them leaching into soil and groundwater if the panel is damaged (e.g. due to reoccurring stress, extreme weather, or poor quality). Examining damaged PV cells in a six-month study, [48] find that both silver and zinc leaching were present, although "drinking water limits for [silver] were only exceeded on a few single days" while zinc levels "did not exceed even the most stringent drinking water limits at any time" (pp. 1674, 1679). Proper maintenance and monitoring can minimise this risk [48].

In the social science literature scholars have highlighted that people may be dispossessed of their land or access to communal resources to make way for solar parks through "legal eviction, illegal expropriation (land grabbing), land speculation," and losing their land rights leading them to become economically displaced [1,156]. In India, large-scale solar parks have created land use conflicts in often poorer, agricultural regions and have seen small landholders dispossessed of farmland, while poor households have also been restricted from accessing biomass and pastoral lands which disproportionately affects lower caste women [110]. Through the concept of 'Cuerpo-territorio' feminist political ecologists underscore the deep connection between land and people, where "[W]hat is experienced by the body is simultaneously experienced by territory in a codependent relationship" [84]. While using different ontological and epistemological frameworks, feminist political ecology reinforces the utility of the concept of solar PV's embodied political ecology because it shows how economic and political decisions about energy shape socio-ecological impacts on marginalised communities.

Stage 4: End-of-life:

Like stages 2–3, the end of life (EoL) stage in the solar PV life cycle is not limited to China, although it outstrips the rest of the world in installed capacity and therefore must grapple with disposing of solar PV on a larger scale. Indeed, the top five markets for cumulative waste volumes are China, the US, Japan, India, and Germany [157]. Meanwhile some developing countries are already reporting environmental and health issues due to poor disposal [119,158]. The embodied political economy of solar PV captures how decisions about disposal contributes to socioecological impacts. Currently, solar PV panels are assumed to last between 25 and 30 years; they can be disposed in landfill, incinerated, re-used, or recycled, and in some cases used for land reclamation [69,159]. LCA studies often exclude the EoL phase [61], although this is changing (e.g. [40,60,63,73,74]; see Table 1). Significant increases in PV waste anticipated after 2030 [157].

To date, most solar PV are sent to landfill or are incinerated [77,95,151,159]. This poses a problem given their toxic materials (e.g. lead, chromium, bismuth, tellurium, cadmium, selenium, nickel, zinc, silver), especially if they are improperly disposed of which could result in toxins leaching into soil and water systems [40,48,63,94,95]. Incineration releases toxic and carcinogenic emissions into the atmosphere [73], impacting human health and ecosystems [94,95].

Recycling is the preferable option [40,44,48]. The dominant c-Si technologies use hazardous materials such as lead and silver, while thin-film PVs such as CdTe uses cadmium/tellurium, which have different recycling possibilities (see Tables 4 and 5). Materials in silicon-based PV

[5] Although toxic materials in PV raises similar concerns (see below).

Table 3
Impacts of Solar PV from Energy Transition Literature.

Source	Life Cycle Stages	Impact Categories	
		Environmental	Social
[29]	Manufacturing	–	Social peripheralization, marginalisation and growing trend of far-right populism stemming from deindustrialisation. May be exacerbated by false promises of solar industrialisation
[99]	Upstream, installation, O&M, EoL	Soil erosion, impacts on vegetation and wildlife; water requirements; GHG emissions; hazardous waste; visual and noise impacts; local climate impacts; land use requirements	–
[100]	Installation, O&M	Ecological degradation (habitat loss, concerns about impacts on local aquifers, dust control issues); land use	Entrenching existing social inequalities (e.g. structural racism); land use contestation
[101]	Installation, O&M	Negative impacts on wildlife (especially birds); deforestation, habitat loss; land use; increased water use impacting local aquifers	Land dispossession; lack of consultation, transparency, and community engagement; devaluation of properties; increased electricity prices
[102]	Installation, O&M	Land use; water requirements	Entrenchment of existing caste, gender, and colonial inequalities
[80]	Upstream	Ground water contamination and air emissions affecting environmental quality	Toxicity, hazardous materials impacting human health/worker exposure
[103]	Installation, O&M	Habitat fragmentation, damage to soils, impacts on wildlife; water use requirements, ground water depletion impacting ecosystems and aquatic wildlife; dust pollution	Land use conflicts; procedural injustices
[22]	Upstream, installation, O&M, EoL	'Embodied ecological injustice' (e.g. environmental damage due to land use for utility-scale solar projects); environmental degradation; pollution	'Embodied labour injustice' (e.g. forced labour in Xinjiang, China); health risks for workers and communities; land conflicts; food insecurity
[104]	–	Carbon intensity associated with solar supply chains	Forced labour accusations in Xinjiang, China; industrial accidents
[105]	Upstream	–	Forced labour accusations in Xinjiang
[106]	Upstream, installation, O&M, EoL	Local waterway contamination, creation of water scarcity; habitat loss, deforestation; noise and dust pollution; resource requirements; hazardous waste often goes into landfill	Human health impacts of silica mining (e.g. silicosis, tuberculosis); livelihood and land dispossession, job losses due to automation. Human health impacts disproportionately impact marginalised and vulnerable groups; procedural injustices; uneven distribution of project benefits; risk of exposure to hazardous chemicals/materials
[19]	Upstream, installation, O&M, EoL	Raw materials requirements; production of waste flows;	Uneven/inequitable access to FiTs; solar PV manufacturing job losses and impacts of deindustrialisation; coal mine closures and nuclear phase out; challenge to centralised supply models; substandard employment conditions/standards; impacting electricity markets in neighbouring countries; disrupting global fossil fuel industry
[107]	O&M	–	Water insecurity/access issues; uneven distribution of benefits (e.g. electricity access, job opportunities) often along the lines of gender and caste; livelihood and land dispossession
[108]	Installation, operation	–	Land use requirements, land and livelihood dispossession, mainly impacting marginalised groups. Uneven development, agrarian economies negatively impacted, promised job creation has not eventuated; uneven access to electricity.
[109]	Resource extraction	Contamination of ecosystems (nearby waterways contaminated with industrial and mining effluents, water pollution, water table depletion, land/soil contamination impacting local animals).	Dispossession of their livelihoods due to automation; energy poverty and food insecurity; workers vulnerable to hazardous and accidents, human health impacts (tuberculosis, respiratory and breathing problems, reported injuries). Dispossessions further entrench existing inequalities.
[110]	Upstream, installation, O&M, EoL	Interspecies inequalities: air, water, soil, and noise pollution, habitat loss, toxic effluents; temporal inequalities: long-term environmental degradation	Demographic inequalities: job insecurity, loss of livelihoods, worker exposure to hazardous materials and practices, resource dislocation, gendered and racial discrimination, use of child labour; spatial inequalities: uneven distribution of waste flows, electricity access, and economic benefits, peripheralization of rural areas, expansion of energy poverty in urban areas; temporal inequalities: further entrenching poverty for marginalised groups,
[111]	Operation	–	Exacerbating local energy poverty; resource and livelihood dispossession/ dislocation; impacts disproportionately impact women

(*continued on next page*)

Table 3 (*continued*)

Source	Life Cycle Stages	Impact Categories	
		Environmental	Social
[112]	Installation, operation	–	Procedural injustices; dispossession of land; impacts disproportionately felt by low status and marginalised groups; land use conflicts
[113]	Installation, operation	–	Increased precariousness for vulnerable groups/ communities; dispossession of land, loss of livelihoods

Table 4
Top Ten Countries for PV Annual and Cumulative Installed Capacity, 2023.

Country	Annual, 2023 (GW)	Country	Cumulative (GW)
China	235.5*	China	662*
European Union	55.8	European Union	268.1
United States	33.2	United States	169.5
India	16.6	India	95.3
Germany	14.3	Japan	91.4
Brazil	11.9	Germany	81.6
Spain	7.7	Spain	37.6
Japan	6.3	Brazil	35.5
Poland	6	Australia	34.6
Italy	5.3	Italy	30.3
Netherlands	4.2	South Korea	27.8

Source: [114, p.10]. Note: European Union includes Germany, Spain, Poland, Italy, and the Netherlands, as well as the other EU member countries. *China's official reporting. The IEA PVPS preliminary assessment indicates this could be 277 GW for 2023 and a cumulative total of 704 GW.

panels are "highly recyclable," such as aluminium, silicon, glass, tin, and lead [73]. Given the dominance of c-Si, Lim et al. [63] note that "treating and recycling EoL crystalline silicon PV are typically given a higher priority" (p. 2). Recycling can reduce environmental impacts when compared with landfill disposal methods (e.g. [63]), although it can be more energy-intensive and increase GHG emissions (e.g. [44]). However, Klugmann-Radziemska and Kuczyńska-Łażewska [60] argue that recycling can help substantially reduce solar PV's *overall* energy requirements and reduce GHG and toxic emissions (see also [63]). Recycling glass can reduce respiratory inorganic impacts [77], while recycling copper and aluminium reduces mining impacts [74]. Recycling solar materials can have negative socioecological impacts. In India for example, recycling is often undertaken by an informal workforce comprised of socially marginalised groups including children, posing ecological and human health and safety risks [110].

Pre-empting discussions on minimising the embodied political ecology of solar PV through extended producer responsibility principles, LCA analysis recognises the module design could also improve EoL management.[6] Currently, solar PV panels have not been designed with the recovery of materials in mind. Consequently, Contreras-Lisperguer et al. [151] argue that PV manufacturers should "consider, *a priori*, designs for non-energy-intensive disassembly and total recyclability of materials, replacing toxic components with non-toxic ones in order to

[6] The OECD defines extended producer responsibility as "an environmental policy approach in which a producer's responsibility for a product is extended to the post-consumer stage of a product's life cycle. An EPR policy is characterised by (1) shifting responsibility (physically and/or economically; fully or partially) upstream towards the producers and away from governments and (2) the provision of incentives to producers to consider environmental considerations when designing their products" [157,p. 20].

Table 5
PV Supply Chain Market Shares and Forced Labour Vulnerabilities, 2020.

Company	Segment (Market Share %)	Forced Labour Exposure
Tongwei	Polysilicon (17), cell (13)	Forced labour exposure in supply chain; no known relationship with XPCC
Daqo	Polysilicon (15)	Polysilicon facility in Xinjiang; forced labour exposure in immediate supply chain; engaged in labour transfers; receives subsidies from XPCC
GCL-Poly	Polysilicon (15), wafer (16)	Polysilicon facilities in Xinjiang; forced labour exposure in immediate supply chain; engaged in labour transfers; direct beneficiary of XPCC
Wacker-Chemie	Polysilicon (13)	Potential forced labour exposure through current/recent contracts with Xinjiang Hoshine Silicon
TBEA/Xinte	Polysilicon (13)	Polysilicon facilities in Xinjiang; possible forced labour exposure in supply chain; engaged in labour transfers; strategic cooperative agreement with XPCC
East Hope	Polysilicon (8)	Polysilicon facility in Xinjiang; engaged in labour transfers; relationship with XPCC unknown
Hemlock Semiconductor	Polysilicon (5)	No known forced labour exposure
OCI	Polysilicon (4)	No known forced labour exposure, although Hoshine Silicon has indicated OCI was a customer
Asia Silicon	Polysilicon (4)	Forced labour exposure in immediate supply chain; relationship with XPCC unknown
Inner Mongolia Dongli	Polysilicon (2)	N/A
LONGi	Wafer (34), cell (11), module (16)	Forced labour exposure in supply chain; power generation plants supported by XPCC
Zhonghuan	Wafer (21)	Shareholder in Xinjiang GCL; forced labour exposure in supply chain; no known relationship with XPCC
JinkoSolar	Wafer (9), cell (6), module (11)	Ingots and wafer manufacturing facilities in Xinjiang; forced labour exposure in immediate supply chain; engaged in labour transfers; no known relationship with XPCC
JA Solar	Wafer (7), cell (6), module (10)	Forced labour exposure in supply chain; power generation plants in XUAR supported by XPCC
Jingyuntong	Wafer (3)	Potential forced labour exposure through current/recent contracts with TBEA/Xinte (through 2021)
Rietech	Wafer (2)	N/A
Solargiga	Wafer (2)	Potential forced labour exposure in supply chain via current/recent contracts with Daqo
Trina Solar	Wafer (2), cell (5), module (8)	Modules manufacturing facility located in Xinjiang (possibly closed); forced labour exposure in supply chain; possible engagement in labour transfers; power generation plants potentially supported by XPCC
Huantai	Wafer (2)	Potential forced labour exposure in supply chain via current/recent contracts with Daqo
Aiko Solar	Cell (8)	N/A
Canadian Solar	Cell (5), module (6)	Forced labour exposure in supply chain; no known relationship with XPCC, although operates a power generation facility in Xinjiang which possibly benefits from relationship with XPCC

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Table 5 (*continued*)

Company	Segment (Market Share %)	Forced Labour Exposure
Jiangsu Zhongyu/ Solarspace	Cell (5)	N/A
Hanwha Q CELLS	Cell (5), module (5)	No known facilities or connections to Xinjiang, although list of suppliers is unknown
Runergy	Cell (4)	Potential forced labour exposure in supply chain via current/recent contracts with LONGi
Risen Energy	Module (4)	Potential forced labour exposure in supply chain; no known relationship with XPCC
First Solar	Module (4)	US-based thin-film producer; not exposed to polysilicon production in Xinjiang, but has self-reported on exposure to forced labour in its supply chain in Malaysia in 2023
Chint	Module (3)	Forced labour exposure in supply chain; no known relationship with XPCC
Wuxi Suntech	Module (2)	No manufacturing facilities located in Xinjiang; forced labour exposure unknown; no known relationship with XPCC (possible subsidies, but unclear)
Other	Polysilicon (4), wafer (2), cell (32), module (31)	–

Sources: [133]: (pp. 28, 34, 41, 48), [105,pp.,48–51], [134]. N/A: not covered in [105]. Note: the Xinjiang Production and Construction Corps (XPCC) is a state-owned paramilitary conglomerate which operates local governments across Xinjiang and a number of publicly listed companies. The XPCC's operations have been linked to internment and re-education camps for minority groups in the XUAR and labour transfer schemes. The XPCC also operates industrial parks in the region which host polysilicon facilities.

Table 6
The use of selected materials in c-Si PV manufacturing.[a]

Material	Main Uses
Aluminium	Module frame; mounting structure; connectors; back contact; inverters
Antimony	Solar-grade glass (used to reduce the long-term impact of untraviolet radiation on the solar performance of glass) and encapsulant (used as a polymerisation catalyst)
Copper	Cables, wires, ribbons, inverters
Glass	Module cover
Indium	Transparent conducting layer (indium tin oxide in silicon heterojunction
Lead	Soldering paste and ribbon coating in c-Si modules
Silicon	c-Si wafers; in the form of high purity quartz, for crucibles to grow monocrystalline silicone ingots via the Czochralski process
Silver	Electronic contacts: silver paste, busbars and soldering
Tin	Solder, ribbon coating in c-Si modules
Zinc	Galvanised steel in mounting structures

Source: [120,p. 20].

[a] Tables 6 and 7 document selected materials for solar PV, other minerals include gallium, nickel, iron, REE and graphite [119,138].

assess the potential to implement up-cycling at a commercial scale" (p. 54) (see also [42]). A preferable option, they argue, is to redesign PV panels and substitute or minimise the use of toxic materials, as well as "incorporating social, economic and environmental benefits to achieve up-cycling with all the materials" [151,p. 56]. Increasing the life span of PV panels/modules will also reduce material requirements [54,73].

In sum, while many of the minerals for solar PV are extracted around the world, processing of minerals and manufacturing of the various component parts of solar PV largely takes place in China and by Chinese companies. This means that many of the decisions that have the biggest socio-ecological impacts are taken by China and Chinese companies. Nevertheless, the global uptake of solar PV has increased dramatically,

Table 7
The use of selected materials in Thin-film CdTe PV manufacturing.

Material	Main Uses
Aluminium	Module frame; mounting structure; connectors; inverters
Antimony	Solar-grade glass (used to reduce the long-term impact of ultraviolet radiation on the solar performance of glass) and encapsulant (used as a polymerisation catalyst)
Cadmium	Absorber layer
Copper	Cables, wires, ribbons, inverters
Glass	Modules cover
Indium	Transparent conducting layer (indium tin oxide)
Molybdenum	Back contact layer
Selenium	Absorber layer in some CdTe cells
Silver	Electronic contacts: silver paste and soldering
Tellurium	Absorber layer (CdTe) and back contact (ZnTe)
Tin	Solder; transparent conducting oxide (indium tin oxide)
Zinc	Galvanised steel in mounting structures; back contact (ZnTe)

Source: [120,p. 20].

with disposal from 2030 also a looming global problem. This section has highlighted the various stages of the solar PV life cycle to demonstrate why and where the socio-ecological impacts are occurring. This better identifies how energy decisions along the life cycle shape the embodied political ecology of producing solar PV for a global market. The following section examines efforts to govern the global solar PV supply chain to minimise its embodied political ecology.

3.3. *Minimising the embodied political ecology in global solar PV chains*

As noted from the outset, voluntary non-state market governance is used to mitigate the socio-ecological impacts of global commodities, although it is generally ineffective in doing so. Governance of solar PV ranges from technical and procurement standards, ecolabels, and industry association standards to more stringent requirements including US legislation to stop solar PV made by forced labour at the border, and an EU Directive on waste. There are therefore a variety of ways in which the embodied political ecology of solar PV is being addressed.[7] However, harder regulations for global supply chains to mitigate socio-ecological impacts remain rare [7,9,10] and may face greater difficulty in being established given tensions between the US and China, particularly around solar PV. This section shows what is being done to address specific impacts from solar PV global supply chains, which only skims the surface of the many decisions that are made in producing solar PV for a global market that contribute to harm. Indeed, as the previous section shows, at each stage of the life cycle there are options for producing a commodity that have fewer negative environmental and social impacts.

The International Electrotechnical Commission (IEC) is a hybrid international organization comprised of national committees of public officials and private representatives. It establishes standards and certification for infrastructure and the international trade of electrical and electronic goods. It has developed guidelines for meeting standards and certification for the performance of solar PV, including IEC 61215 for c-Si PV. While manufacturing will remain in China for the near future, France and South Korea have adopted standards and procurement processes to reduce the emissions intensity of solar PV manufacturing. In the United States, a 'voluntary procurement standard' was anticipated for 2023, with benchmarks for GHG emissions to meet 'low carbon' and 'ultra-low carbon' certifications for PV modules. This is being undertaken by the Global Electronics Council, which is a US non-government organization. Its EPEAT registry provides third-party verification and ecolabelling for solar PV products throughout their life cycle including module recycling and Energy Star manufacturing equipment [104,161].

In the US, the Solar Energy Industries Association (SEIA) has looked

[7] The United Nations is trying to establish traceability protocols for critical minerals to mitigate socio-ecological impacts [160].

to address forced labour concerns by recommending divestment from Xinjiang and developing traceability protocols in the supply chain. SEIA has also received pledges from approximately 250 manufacturers to increase transparency in their supply chains and vulnerabilities to forced labour [104,105]. In 2023, the SEIA became an 'accredited standards development organisation,' following approval from the American National Standards Institute. In September 2024, the SEIA [162] released its Solar and Energy Storage Supply Chain Traceability Standard 101 for public comment. Alternatively, the Solar Stewardship Initiative, that emerged out of Europe, is a multistakeholder initiative comprised of solar manufacturers and buyers, civil society, and trade associations. It has 40 members and is a "solar-specific sustainability assurance scheme based on independent third-party verification" [163], In late 2023 they created an ESG standard for the solar industry and have just released a traceability standard [163].

Stringent rules have been applied to solar PV commodities. When the forced labour allegations came to light the US announced an executive action in June 2021 for US Customs and Border Protection to stop silica-based materials entering at the border if they could not provide evidence of their origin. Shortly after, the US passed the Uyghur Forced Labor Prevention Act (the UFLP Act) which became law in December 2021 [133]. The Act, which came into effect from 21 June 2022, has banned goods produced in Xinjiang [125] (see below). China has in turn responded to these changes by shifting polysilicon production to Inner Mongolia [125]. US efforts may shift towards 'reshoring' (increasing domestic production capacity of certain parts of the supply chain) and shifting away from c-Si PV technologies towards thin-film and third-generation PV technologies to avoid the many of the socio-ecological impacts of the current global supply chain [104,105]. Greater transparency in Chinese governance of the solar industry is needed to understand how negative socio-ecological impacts are being managed and mitigated.

While the US has led efforts to tackle forced labour in the global solar PV supply chain, the EU is at the forefront of reducing the impact of waste at the end-of-life stage. The rate of recycling varies by region and country, with the number of PV modules being recycled in the EU reaching almost 95 % [119], compared with below 10 % in the US [119,164]. The US has no specific regulations for solar PV waste/recycling. However, some regulatory frameworks for PV recycling have been developed at the county level (see [165]). China, the US, India, Japan and Australia "still lack specific regulations for managing EoL PV modules, which are often treated under a general regulatory framework for hazardous or non-hazardous solid waste" ([119,p. 117]; see also [159,166]). China, despite dominating solar PV manufacturing and installed capacity, "has no proposals for the dumping of the old panels" [86,p. 4], although it is investigating recycling [95]. Around the world voluntary schemes for PV recycling do exist, although they are not considered to be profitable and mainly have reputational benefits [119].

The EU was the first to establish a regulatory framework for solar PV waste in the form of the 2012 *Waste Electrical and Electronic Equipment (WEEE) Directive (2012/19/EU)*. As [119] notes:

> The European framework follows an extended-producer-responsibility approach, making producers or sellers who put PV modules on the EU market liable for the costs of collection, handling and treatment of their products, either through their own takeback and recycling programme or through producer compliance schemes (p. 117).

These regulations apply to *any* producer supplying the EU market, wherever they are based [157]. Coming into effect in 2014, it has been implemented differently by states, and some have clearer standards [167].

Examining the status of waste management of solar PV in developing countries, [95] note that despite an awareness of the impacts, "nothing is being done presently to recycle or plan for the management of the items at end-of-life (EoL)" (p. 1). This poses ethical problems as a significant share of WEEE produced in developed countries ends up in developing states, sometimes as part of 'humanitarian aid,' while other times it is shipped illegally. Developing states electronic waste recycling methods usually use manual processes in informal settings, creating environmental and human toxicity hazards. Between 50 and 80 % of e-waste produced in developed countries goes to developing countries for continued use, which often contravenes international law [95]. As they state:

> It is unethical for governments and key players in the PV industry to neglect or violate the Basel Convention on Transboundary Movement of Hazardous Wastes and Their Disposal 1989. The convention is the international policy that regulates the movement of these harmful wastes within member nations that signed it. It is obvious that most, if not all, developing countries presently do not have specific laws or focal regulations dealing with the management of PV wastes (p. 6).

The EU's WEEE Directive seeks to address this in Europe [86].

To summarize, there are a few voluntary transnational standards emerging for the global solar industry (and for extracting critical minerals for RE, see [10]). These are increasingly requiring traceability and independent verification to determine whether socio-ecological impacts have been minimised. However, the voluntary nature of initiatives tends to undermine their efficacy. Further transparency of sustainability and justice regulations in China is also needed to improve the overall embodied political ecology of the global solar PV supply chain, given China and Chinese companies dominance in processing and manufacturing. Attempts by the US to stop forced labour by preventing solar PV at the border without proof of origin may not necessarily have ended the abuse. It remains unclear how to rachet up measures to reduce the embodied political ecology of solar PV given the urgency of the climate crisis while solar PV prices do not incorporate socio-ecological impacts. While the EU is advancing efforts to establish extended producer responsibility principles that minimise some of the socio-ecological impacts, this has yet to be taken up in other large markets such as the US. While geopolitical competition could encourage the uptake of new generation solar PV technologies with fewer impacts and could facilitate 'reshoring' of manufacturing also with fewer impacts, increasing tensions between the US, China and the EU do undermine international cooperation and the likelihood of global agreement around strengthening global supply chain governance (through for example traceability protocols). For now, piecemeal attempts to reduce the embodied political ecology of solar PV remain.

4. Conclusion

Producing commodities through global supply chains for global markets creates a myriad of socio-ecological impacts which may not be widely known. We conceptualize solar PV as having an embodied political ecology to recognise that energy decisions along the global supply chain contribute to harm. Embodied political ecology encompasses the idea that every solar PV may have been produced through extractivist violence, dispossession, and forced labour, and may contribute to eco-toxicity, pollution, and water depletion, among other impacts. We have attempted to document all of the known socio-ecological impacts of producing solar PV and locate where the impacts are occurring along the global supply chain. This is important for capturing the full range of impacts compared with site specific case studies and inductive analyses that dominate the social science literature on the renewable energy transition to date. It provides a way to locate current case study research of the impacts on solar PV within a breadth of socioecological impacts.

Using life-cycle analysis (LCA) and non-LCA studies, we attempted to document all impacts along the life cycle of solar PV from extraction, processing and manufacturing; to installation; operation and maintenance; and to the end of life. Demonstrating where the impacts are occurring throughout the life-cycle of the product provides detailed understanding of the political and economic decisions being made. For

example, technical decisions have and are being made about the type of panels produced, their composition, and their disposal all of which have significant impacts that could be reduced. Indeed, the LCA literature provides recommendations for how to minimise these impacts at each stage of the life cycle through improving solar PV design for recycling; to improved sizing and siting for installation, operation and maintenance; and the best means of disposal. Political questions about decision making, democratisation, community control and ownership are also needed given that these that all shape the socio-ecological impacts of solar PV.

We then mapped these impacts onto the global supply chain. Notably, the biggest socioecological impacts occur in the extraction, processing and manufacturing stages of solar PV, with GHG emissions, acidification, and ozone depletion impacts linked to the fuel mix used in China and in Xinjiang specifically. In documenting the embodied political ecology of the commodity, we highlight the need for China and Chinese companies to enact greater and more transparent sustainability and justice measures in the extraction, processing and manufacturing of all module components stages, particularly regarding forced labor. Nevertheless, socioecological impacts occur along the global supply chain, in many jurisdictions, where different energy decisions can minimise solar PV's embodied political ecology.

Given the rapidity of the uptake of solar PV, there are concerns that the embodied political ecology of the commodity will continue to be overlooked. This is because the urgency of the climate crisis and dependence on China has policy makers concerned about future projections of solar PV given increasing tensions between the US and China and increasing trade protectionism. While there are efforts to manage solar PV's embodied political ecology through voluntary non-state market governance initiatives, US laws on forced labor, and the EU's waste directive, these are piecemeal. A more comprehensive means of regulating the embodied political ecology of solar PV is needed. Whether this is enough to help propel a broader sustainability transformation requires further research.

CRediT authorship contribution statement

Susan Park: Writing – original draft, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Oliver Summerfield-Ryan:** Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known conflicts of interest.

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Data availability

Data will be made available on request.

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